

HYBRID TEMPORAL FORECASTER

From Baselines to Regime-Gated
Hybrid Architectures & Failure Analysis

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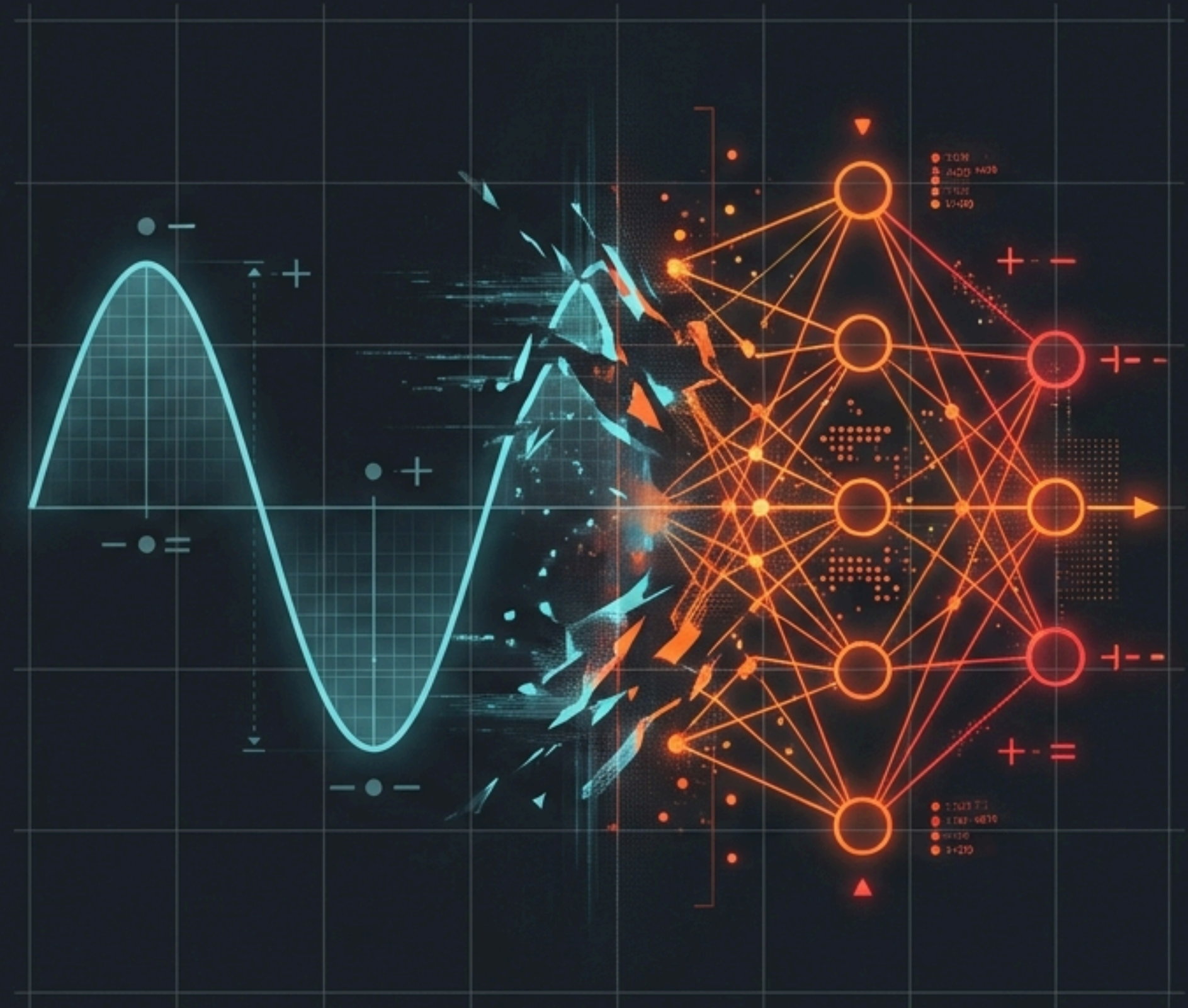
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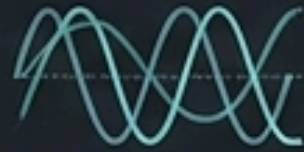
Non-Stationary Volatility in Grid-Level Forecasting

High-Stakes Accuracy



Precise load forecasting balances supply, minimizes costs, and prevents cascading blackouts.

Complex Temporal Dynamics



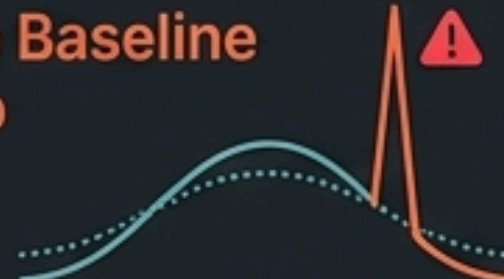
Demand exhibits multiple overlapping, cyclical seasonalities across hourly, daily, and yearly horizons.

Non-Stationarity & Volatility



Exogenous shocks like unpredictable weather trigger sudden regime shifts in energy consumption.

The Baseline Gap



Traditional mathematics aggressively smooth over volatility, entirely missing critical, high-frequency spikes.



Theoretical Demand

Real-World Grid Volatility

Analyzing Temporal Signatures in Real Energy Demand



Continuous High-Resolution Data

Multi-year sequence of continuous Real Energy Demand (MW) recordings.

Inherent Periodicity

Strong temporal signatures tracking specific hours, days of the week, and months.

Extreme Variance Boundary

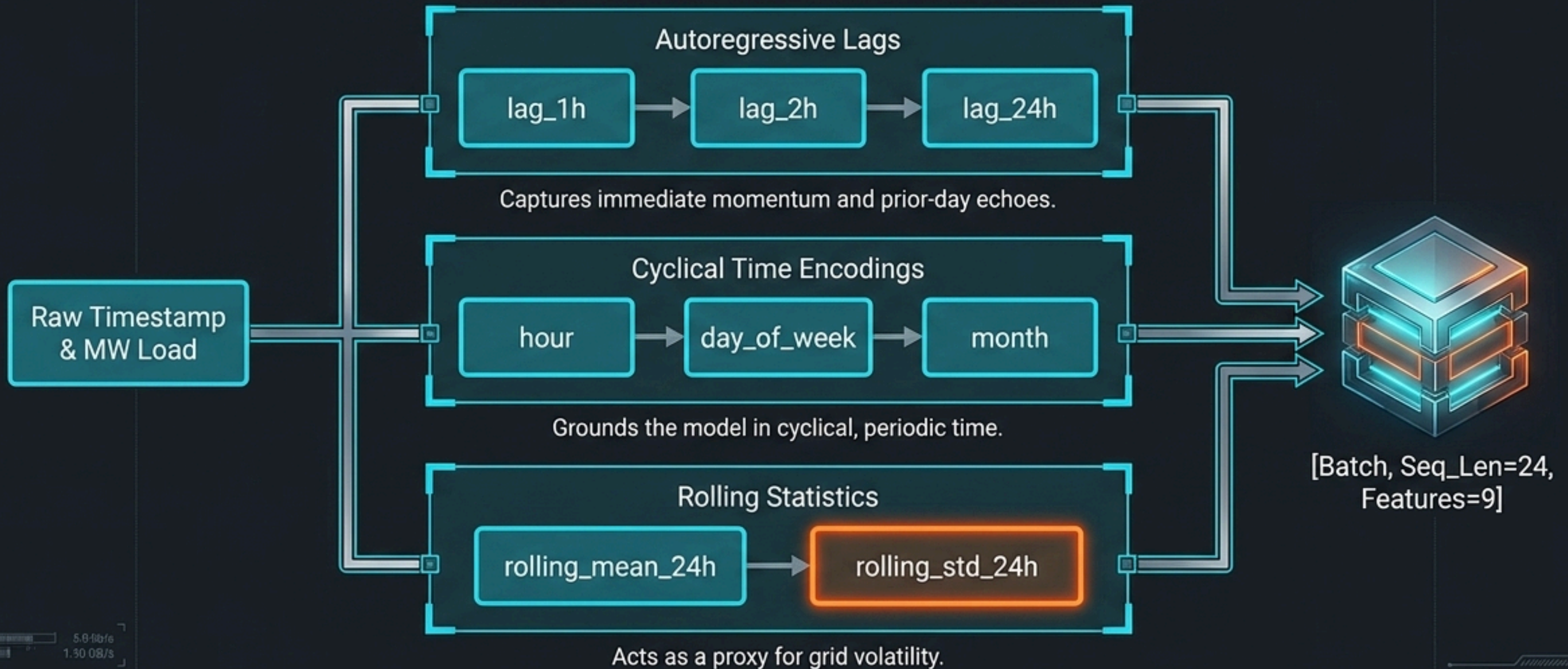
Observations span from stable base-load minimums (~4,700 MW) to massive peak loads (>36,000 MW).

Objective

Sequence-to-one forecasting predicting the next-step Real Energy Demand based on historical windows.

Engineering the Temporal Context

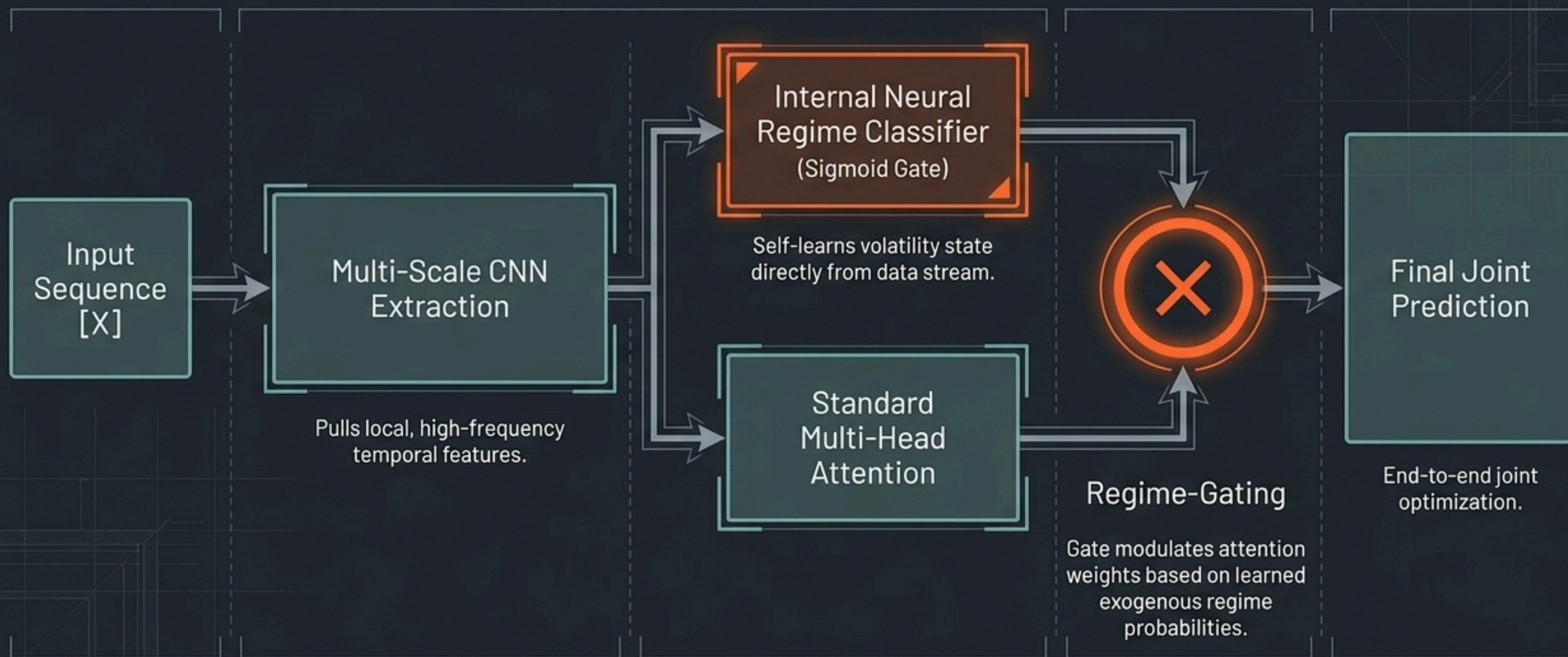
Sequence Generation Pipeline



Architectural Evolution: Overcoming Baseline Limitations

	SVR (Baseline 1)	Time-Series Transformer (Baseline 2)	Advanced Hybrid Forecaster (Novel)
Core Mechanism	RBF Kernel mapping	Self-Attention with Positional Encoding	CNN extraction with state-aware, regime-gated attention
Strengths	✓ Fast and mathematically robust on baseloads	✓ Captures long-term sequence dependencies	✓ End-to-end deep temporal network uniting local features and dynamic state-awareness
Critical Weakness	✗ Heavily penalizes variance; artificially smoothed, lagging forecasts	✗ Suffers from severe high-frequency noise during stable regimes	✗ Computationally expensive scaling

The Regime-Gated Attention (RGaA) Mechanism



Advanced Regularization & Curriculum Strategies

Volatility-Based Curriculum

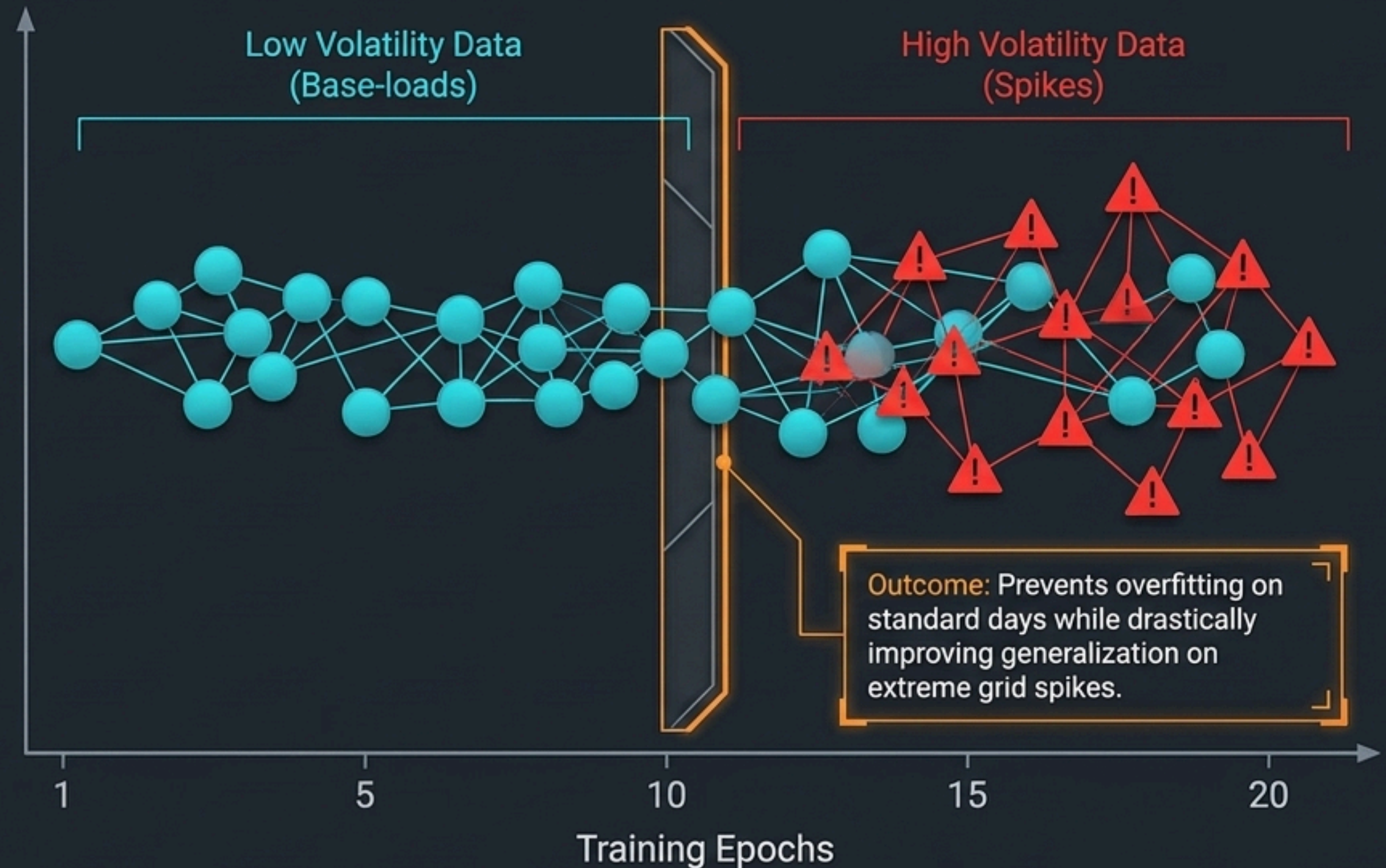
Data split into "Easy" (normal) and "Hard" (extreme) batches using the engineered `rolling_std_24h` feature.

Progressive Difficulty Loading

Model mathematically stabilizes on base-loads first before weights are exposed to highly volatile anomalies.

Time-Series MixUp

Linear interpolation of sequences ($\alpha=0.2$) synthetically generates novel transitional training examples.

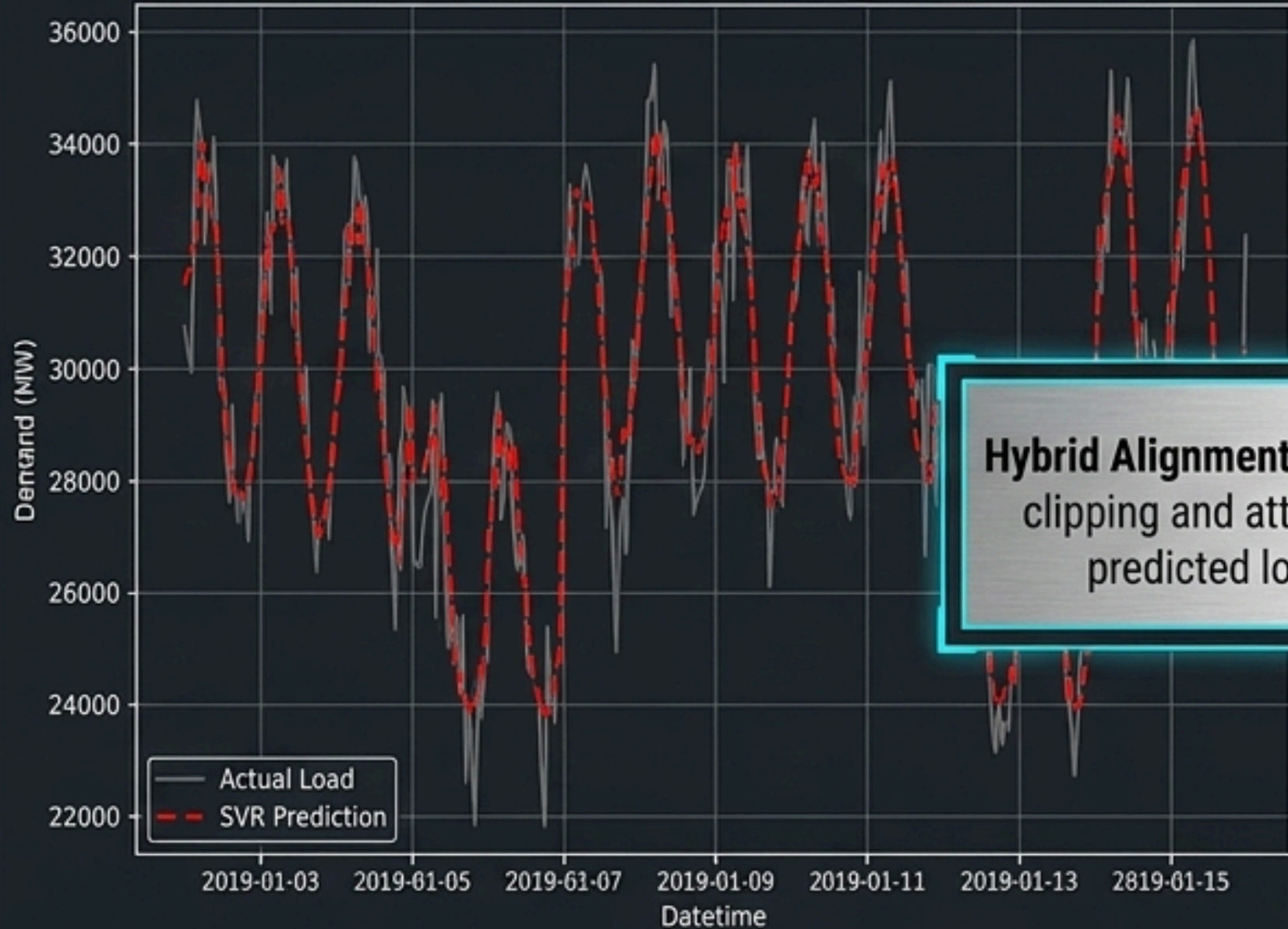


Comparative Validation & Predictive Superiority

[Architecture]	[MAE Tracking]	[MSE Tracking]	[Robustness Profile]
SVR (Baseline)	High Error	Phase Lagging	Systematically underestimates peak demand amplitudes.
Standard Transformer	Moderate Error	Amplitude Catching	Tracks peaks effectively but suffers false-positive volatility during stable regimes.
Regime-Gated Hybrid	Superior Low Error	Aggressive Tracking	Maintains predictive integrity over multi-week holdouts. Regime-gating smooths base-loads while reacting precisely to true spikes.

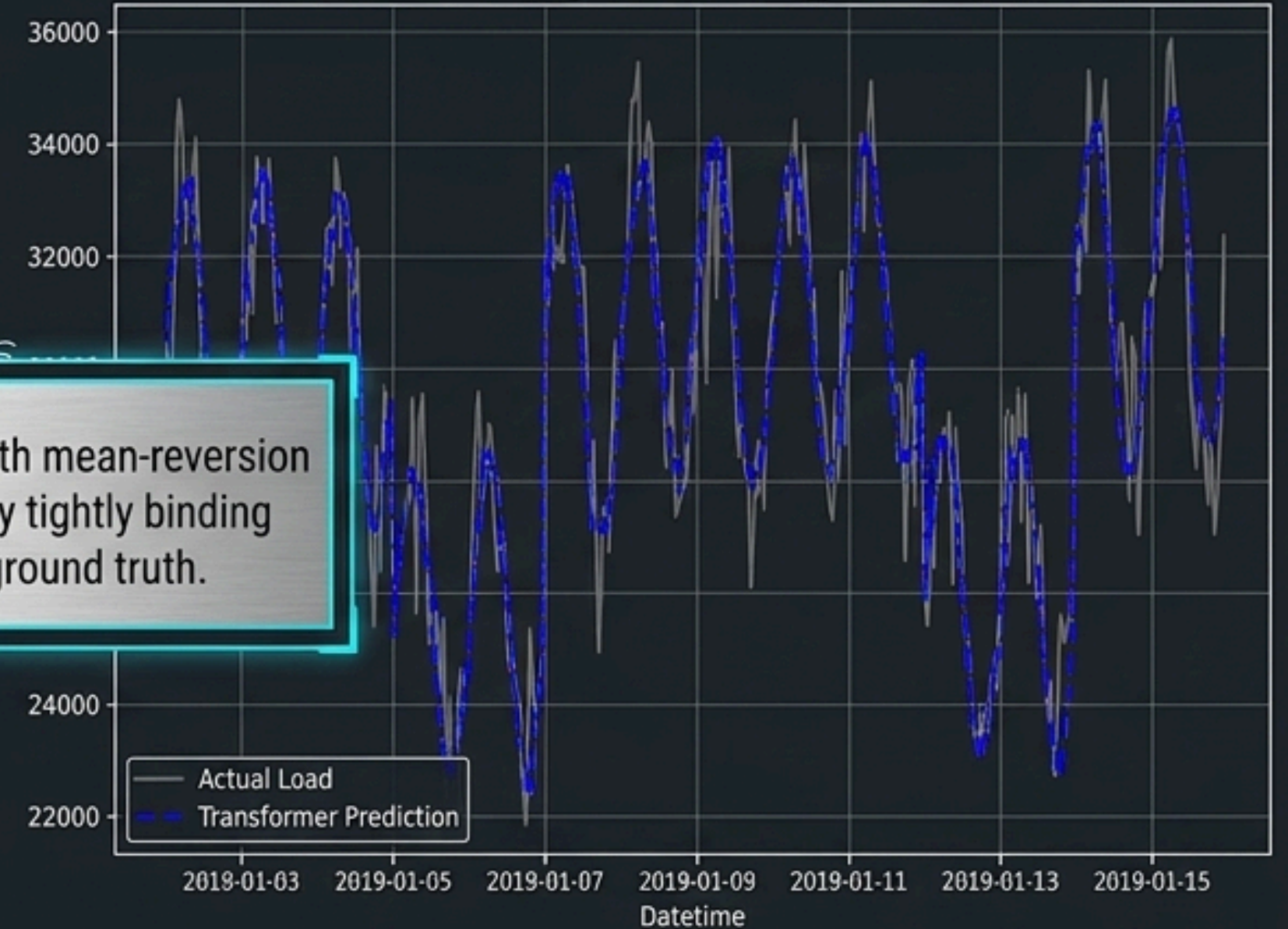
Visualizing the Predictive Edge

SVR Baseline (RBF Kernel) vs Actual Energy Demand (2 Weeks)



Hybrid Alignment: Resolves both mean-reversion clipping and attention-jitter by tightly binding predicted load to actual ground truth.

Transformer Instability: vs Actual Energy Demand (2 Weeks)



SVR Mean-Reversion: RBF kernel aggressively minimizes error by reverting to the mean, physically clipping peak demand amplitude.

Transformer Instability: Sequence-to-sequence attention successfully catches amplitude frequencies, but catastrophically overfits to noise.

Mathematical Boundaries & Model Breakage

[Rigorous Autopsy Protocol]

Actively forcing the model to fail to map its absolute mathematical limits.

[Derivative Velocity Threshold]

Total breakdown occurs when the first derivative (rate of change) abruptly exceeds the historical volatility proxy.

$$\frac{\Delta \text{Load}}{\Delta t} > \text{rolling_std_24h} = \text{Predictive Failure}$$

[The Exogenous Shock Disconnect]

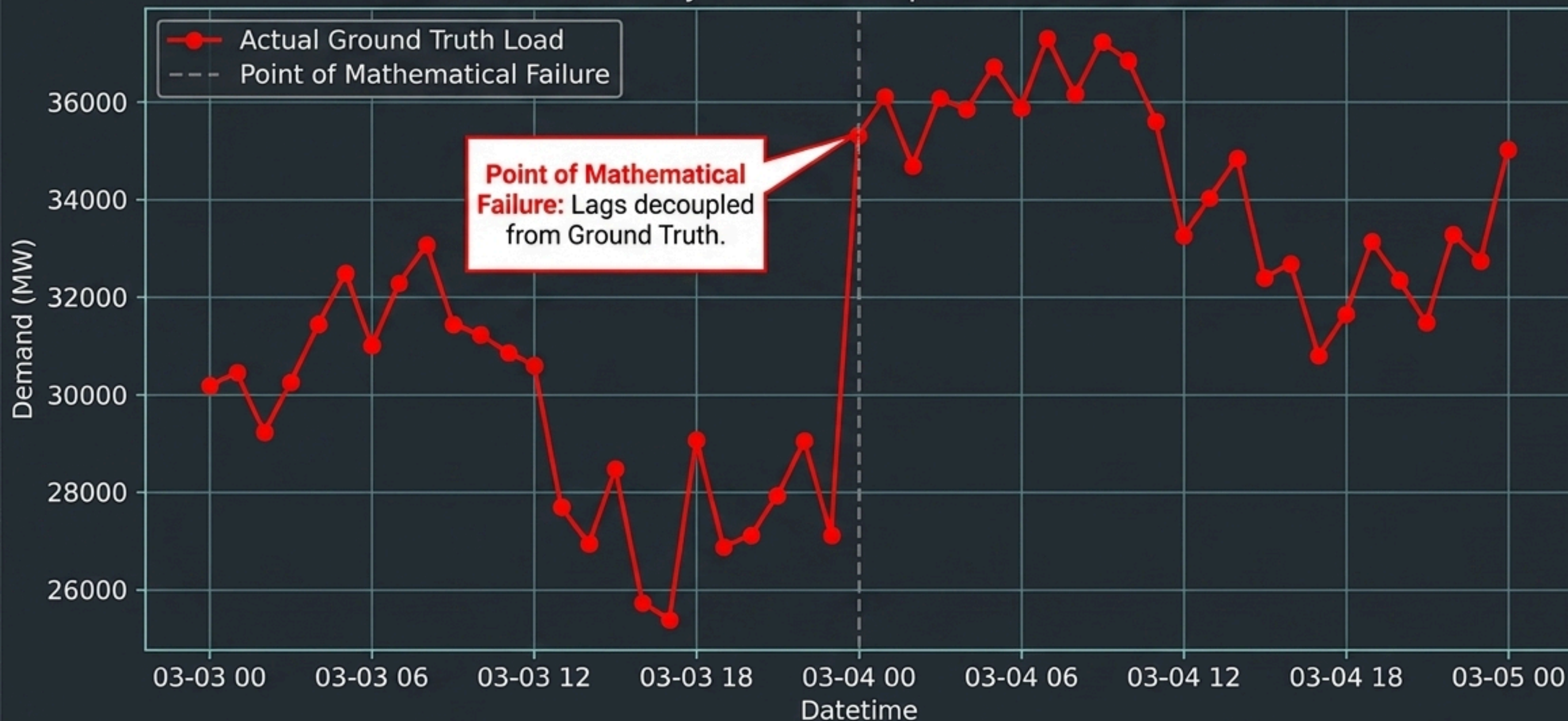
Purely autoregressive models are mathematically blind to spikes driven by unrecorded external variables.

[Regime Shift Lag]

The gating mechanism requires exactly one sequence step (seq_length=24) to recalibrate to a newly established volatile regime.

Autopsy of an Exogenous Shock Failure

Zoomed Failure Analysis: Sudden Spike at 2019-03-04 00:00:00



Ground Truth Anomaly: Actual grid load jumps instantaneously from ~27,000 MW to ~35,000 MW on 2019-03-04.

Math Breakdown: At the exact inflection point, historical lags provide zero predictive prior for the amplitude.

The Re-calibration Window: Architecture mandates a brief sequence of time steps post-spike to absorb the new regime.

Current Architectural Limitations

Feature Blindness

The model is strictly autoregressive. It mathematically lacks critical external causal vectors such as localized weather shifts, grid outages, and holidays.

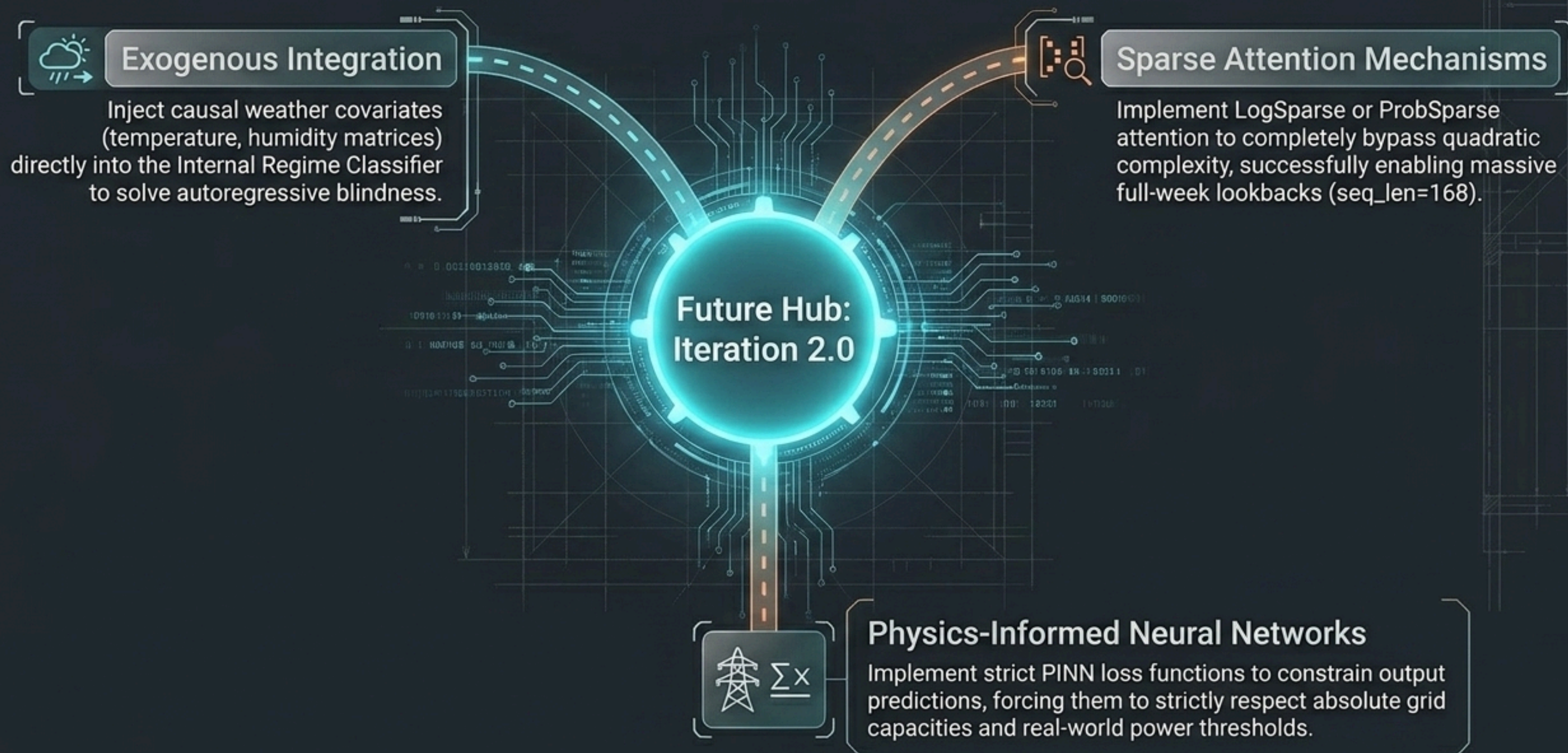
Quadratic Computational Overhead

Regime-Gated Multi-Head Attention scales quadratically with sequence length, strictly bottlenecking massive historical lookback windows.

Synthetic Artifacts

Linear interpolation via Time-Series MixUp can occasionally synthesize biologically/physically impossible grid behaviors during the training loop.

Next-Generation Forecasting Trajectories



Summary of Research Contributions

Developed an end-to-end **Regime-Gated Hybrid architecture** that fundamentally **outperforms** baseline **Transformer** and **SVR** models

Successfully mapped temporal volatility directly to the training loop via **Volatility-Based Curriculum Learning** and **Time-Series MixUp**

Executed rigorous mathematical **failure analysis**, proving autoregressive architectures **fundamentally fracture during unrecorded exogenous shocks**

Established a mathematically **robust**, extensible framework for next-generation, **grid-aware** energy forecasting

Defense Concluded
Floor Open for Examination